

Generation Interconnection System Impact Study Report

For Queue Project Q18-1038

Yabucoa 100 kV
February 18, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1038. The Interconnection Customer proposes a 98.4 MW solar photovoltaic facility interconnecting at the Yabucoa 100 kV substation in the Illinois area. The request was received on August 13, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 339 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,955,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Yabucoa-Saticoy 100 KV	Yabucoa-Lucas 100 KV	109.9	118	61.9
Darlington Road-Wayne Trace 100 KV	Richland-Dekalb Substation 500 KV	109.8	556	5.9
Baraga Vlg 500-230 kV Transformer	Trego-Beaver 500 KV	105.8	1,991	13.0
Newport ie-Post Rock 230 KV	Trego-Beaver Substation 500 KV Line	104.9	2,343	13.1
Myrtle Point-Post Rock 230 KV	Trego-Beaver 500 KV	104.2	2,391	12.6
Mason St 500/230kV Transformer Bank	Trego-Beaver 500 KV	104.1	2,151	11.8
New Departure-Adams Substation 230 KV line	Trego-Beaver 500 KV	103.6	1,433	7.7
Old Forge-Involta 230 KV	Hylebos-Ravenna 500 KV	103.5	520	5.2
Hillcrest-Segs 5 100 KV	Richland-Dekalb 500 KV line	102.9	710	7.1
Stoner Substation-Ledges 230 KV	El Paso-Ravenna 500 KV	102.7	618	5.7
Norwich-Larson Street 230 KV L	Nowich-Barrett 500 KV	102.6	1,633	8.7
Beaver-Onancok Diesel Generation Facility 230 KV	Beaver-Eufaula 500 KV	102.4	2,230	10.7
Minnehaha-Sidney A Murray Jr Hydroelectric 230 KV	Richland-Dekalb 500 KV Line	102.3	1,403	16.3
George Neal South-Zona Libre 6007 230 KV	Galion Generating Sta-Trego 500 KV	102.3	1,196	6.4
Slocum Rd-Mcneil 230 KV	Richland Substation-Dekalb 500 KV L	102.2	2,032	25.4
Slocum Rd-Dekalb 230 KV	Dekalb-Richland 500 KV	102.0	2,066	25.4
Highgate-Herald-Twin Cities 230 KV L	Mason St 500/230 kV XFMR	102.0	1,416	6.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Zoa Libre 6007-Kampsville 230 KV	Trego-Galion Generating Sta 500 KV line	101.9	1,226	6.4
Breed-Mcneil 230 KV	Richland-Dekalb 500 KV	101.7	2,060	25.4
Sidney A Murray Jr Hydroelectric-Breed Substation 230 KV	Richland-Dekalb Substation 500 KV	101.7	2,060	25.4
Moreno Tap-Kampsville 230 KV	Galion Generating Station-Trego 500 KV	101.7	1,243	6.4
Atascocita-North Cambridge 230 KV	El Paso-Ravenna 500 KV	101.5	1,216	8.9
E Methuen-Madeline 230 KV L	Baraga Vlg-Gray'S Flat 230 KV	101.1	1,864	9.4
Adirondack 500 kV/230 kV Transformer	E Methuen 500 kV/230 kV XFMR	101.1	1,694	5.0
Norwich 500 kV/1 kV Autotransformer	Norwich-Barrett 500 KV Line	100.3	1,120	6.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yabucoa-Saticoy 100 KV	Reconductor line	\$416,000	36% (\$152,000)
Darlington Road-Wayne Trace 100 KV	Reconductor line	\$557,000	1% (\$4,000)
Baraga Vlg 500 kV/230 kV TX	Replace transformer with higher-rated unit	\$20,186,000	1% (\$118,000)
Newport ie-Post Rock 230 KV	Reconductor line	\$14,942,000	1% (\$91,000)
Post Rock-Myrtle Pt 230 KV L	Reconductor line	\$6,954,000	1% (\$42,000)
Mason St 500/230kV Transformer Bank	Replace transformer with higher-rated unit	\$21,314,000	1% (\$134,000)
New Departure-Adams 230 KV	Reconductor line	\$11,493,000	1% (\$72,000)
Involta-Old Forge Substation 230 KV	Reconductor line	\$4,212,000	1% (\$28,000)
Segs 5-Hillcrest Substation 100 KV	Reconductor line	\$1,596,000	1% (\$11,000)
Stoner Substation-Ledges 230 KV	Reconductor line	\$6,694,000	1% (\$41,000)
Norwich-Larson Street 230 KV L	Reconductor line	\$5,448,000	1% (\$32,000)
Beaver-Onancok Diesel Generation Facility 230 KV	Reconductor line	\$8,697,000	1% (\$53,000)
Minnehaha-Sidney A Murray Jr Hydroelectric 230 KV	Reconductor line	\$12,138,000	1% (\$83,000)
Zona Libre 6007-George Neal South 230 KV	Reconductor line	\$8,827,000	1% (\$53,000)
Slocum Rd-Mcneil 230 KV	Reconductor line	\$3,937,000	1% (\$27,000)
Slocum Rd-Dekalb 230 KV	Reconductor line	\$16,014,000	1% (\$109,000)
Highgate-Herald-Twin Cities 230 KV L	Reconductor line	\$10,432,000	1% (\$70,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Zona Libre 6007 Substation-Kampsville 230 KV Line	Reconductor line	\$9,455,000	1% (\$57,000)
Breed-Mcneil 230 KV	Reconductor line	\$8,157,000	1% (\$55,000)
Sidney A Murray Jr Hydroelectric-Breed Substation 230 KV	Reconductor line	\$16,241,000	1% (\$110,000)
Moreno Tap-Kampsville 230 KV	Reconductor line	\$20,596,000	1% (\$125,000)
Atascocita-North Cambridge 230 KV	Reconductor line	\$6,463,000	1% (\$39,000)
E Methuen-Madeline 230 KV L	Reconductor line	\$9,948,000	1% (\$57,000)
Adirondack 500 kV/230 kV Transformer	Replace transformer with higher-rated unit	\$34,465,000	0% (\$183,000)
Norwich 500-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$35,450,000	1% (\$209,000)
	Total allocated to project		\$1,955,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.